

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

The limited availability of annotated data in specialized domains, particularly medical imaging, constitutes a persistent impediment to the training of robust and transferable vision models. Although self-supervised masked image modeling (MIM) provides an attractive pretraining paradigm, dominant instantiations continue to depend on random, high-ratio masking schemes that are computationally inefficient and only weakly coupled to semantic content; moreover, existing region-aware variants frequently rely on reconstruction heuristics or supervised cues, thereby constraining their adaptability across tasks, anatomical targets, and imaging modalities. We introduce Mask What Matters, a controllable text-guided masking framework for self-supervised medical image analysis. By exploiting vision-language models for prompt-conditioned region localization, the proposed method applies differentiated masking in a flexible manner, intensifying reconstruction pressure over diagnostically salient regions while attenuating redundant supervision in background areas. This controllable formulation promotes semantic alignment, strengthens representation learning, and improves generalization across heterogeneous downstream settings. Comprehensive evaluation over multiple medical imaging modalities, including brain MRI, chest CT, and lung X-ray, indicates that Mask What Matters consistently surpasses existing MIM methods (e.g., SparK), yielding improvements of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection. Notably, these gains are obtained under substantially lower overall masking ratios (e.g., 40% vs. 70%), suggesting that semantic selectivity can compensate for, and even outperform, indiscriminate high-ratio corruption. Together, these findings demonstrate that controllable, text-driven masking offers a viable route toward semantically aligned self-supervised learning and advances the construction of robust vision models for medical image analysis.

1 INTRODUCTION

Image classification, segmentation, and object detection constitute foundational problem settings in computer vision, and recent progress in these areas has been driven predominantly by machine learning, especially deep learning. Yet the empirical success of high-capacity deep models is typically predicated on access to large-scale annotated datasets, a prerequisite that becomes particularly restrictive in specialized domains such as medical image processing, where expert annotations are scarce, costly, and time-consuming to obtain, thereby impeding the broader deployment of deep learning systems.

Self-supervised learning (SSL) addresses this annotation bottleneck by using unlabeled data for pretraining, enabling models to acquire transferable feature representations without manual supervision (Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)). Within the broader SSL landscape, MIM has emerged as a prominent paradigm: by masking portions of an image and requiring reconstruction of the occluded content from visible context, it induces the model to internalize both structural regularities and semantic dependencies. This strategy has achieved

strong performance across diverse downstream tasks and has been successfully adapted to medical image analysis [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

However, prevailing MIM approaches, including masked autoencoders (MAE) [He et al. (2022)] and SparK [Tian et al. (2023)], typically employ random masking strategies that are not explicitly aligned with the regions of interest required by downstream tasks. Such task-agnostic corruption introduces two principal limitations: (1) the resulting representations may become semantically misaligned with downstream objectives, thereby weakening transfer performance; and (2) in order to increase the probability that informative image regions are masked, existing methods often adopt extremely high masking ratios (e.g., 75% in MAE), which intensifies reconstruction difficulty and increases the demand for large-scale data and computational resources. In contrast, natural language exhibits compact structure and high semantic density, permitting effective learning even at comparatively low masking ratios [Devlin et al. (2019)]. The dependence of image-based MIM on high masking ratios thus stems largely from intrinsic visual redundancy and from the absence of semantic-aware masking mechanisms capable of directing model capacity toward genuinely critical regions.

This problem is especially acute in medical imaging, where semantic sparsity is a defining characteristic: diagnostically relevant evidence is frequently concentrated in small, localized structures, such as lesions or organs, whereas extensive portions of the image may consist of background with limited task-specific information. Furthermore, medical imaging encompasses heterogeneous modalities, including MRI, CT, and X-ray, whose task-relevant regions vary substantially in shape, scale, and spatial distribution across clinical targets. These properties impose unusually stringent requirements on the efficiency, controllability, and cross-modal generalizability of masking strategies.

To address these challenges, we propose a controllable, text-guided masking framework, hereafter denoted **Mask-What-Matters (MWM)**. Given a user-specified open-vocabulary description or phrase, a pretrained vision-language model produces zero-shot localization cues, which are transformed into robust regions of interest (ROIs). MWM subsequently applies region-specific, prompt-conditioned masking by assigning higher masking ratios to semantically important areas (e.g., lesions/organs) and lower ratios to background, thereby embedding downstream task semantics directly into the pretraining signal without requiring per-image reports or labels.

In summary, the principal contributions of this work are as follows:

- We present the first controllable text-guided masking framework for medical imaging. MWM integrates vision-language models (BiomedCLIP) with segmentation refinement (SAM) to localize task-relevant regions from open-vocabulary prompts and to impose differentiated ROI-versus-background masking, thereby mitigating the semantic misalignment inherent in random masking.
- We formulate an annotation-free and backbone-agnostic design. MWM requires neither per-image reports nor labels and can be incorporated, under a unified protocol, into both ViT- and ConvNet-based MIM pipelines.
- We demonstrate consistent improvements across modalities and tasks. On brain MRI, chest CT, and X-ray datasets, MWM surpasses state-of-the-art MIM baselines (e.g., SparK), improves classification, detection, and segmentation performance, and substantiates the promise of text-driven generative pretraining for medical imaging.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling first gained prominence in natural language processing through bidirectional encoder representations from transformers (BERT) [Devlin et al. (2019)], which learns contextual representations by masking tokens and predicting them during pretraining. Motivated by this principle, [He et al. (2022)] introduced MIM to computer vision, where image regions are occluded during pretraining and the model is optimized to reconstruct the missing content. This paradigm has exhibited strong generalization in both natural-image and medical-image domains [Gupta et al. (2025); Liu et al. (2021); Huang et al. (2023)]. Representative methods include MAE [He et al. (2022)], MedMAE [Gupta et al. (2025); Xiao et al. (2023)], and SparK [Tian et al. (2023)]: MAE relies on high-ratio random masking to encourage ViTs to emphasize high-level semantic abstractions; MedMAE extends this formulation to medical images through domain-specific adaptations; and SparK transfers

the strategy to convolutional backbones, exploiting their intrinsic multi-scale features to improve pretraining efficacy.

Despite these advances, conventional MIM approaches generally adopt random or structure-agnostic masking strategies that disregard the semantic requirements imposed by downstream tasks. This non-selective corruption can cause the model to allocate representational capacity to irrelevant regions during pre-training, thereby reducing transfer effectiveness. In response, recent studies have explored region-aware masking strategies intended to improve semantic alignment, with two major directions emerging.

One line of work develops loss-driven adaptive masking, in which methods such as hard patches mining (HPM) [Wang et al., 2023a] and its extension AnatoMask [Li et al., 2024] dynamically analyze regional reconstruction errors and assign higher masking probabilities to areas with larger losses, with the goal of prioritizing semantically valuable structures. However, because reconstruction difficulty is not necessarily equivalent to task relevance, this strategy may redirect the model toward hard-to-reconstruct but clinically irrelevant regions; in brain magnetic resonance imaging (MRI), for instance, complex structures such as the spinal cord or eyeballs may receive disproportionate attention despite being unrelated to downstream objectives such as tumor detection.

Another line of work incorporates external perceptual modules as masking guides. FocusMAE [Basu et al., 2024], for example, leverages pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking; although this design can be effective in particular tasks, it requires supervised RPN training on domain-specific annotated data, which limits cross-task generalization.

Collectively, these limitations motivate a masking mechanism that is simultaneously more flexible, semantically aligned, and generalizable. Natural language, by providing a rich and interpretable source of weak supervision, offers a promising basis for guiding masked modeling in a task-aware yet annotation-free manner.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

The rapid development of vision-language alignment models has made it increasingly feasible to perceive and identify task-relevant regions under zero-shot and open-world conditions [Radford et al., 2021]. Canonical approaches in the Contrastive Language-Image Pretraining (CLIP) family learn a unified embedding space for images and text through large-scale joint pretraining on image-text pairs and have consequently been adopted widely for zero-shot and open-set visual recognition. In medical imaging, domain-adapted extensions such as BiomedCLIP [Zhang et al., 2023] further improve cross-modal understanding and attain robust performance across diverse downstream applications. Although vision-language pretrained models have been incorporated extensively into supervised learning pipelines, existing studies primarily emphasize downstream applications, including object detection and semantic segmentation [Zhong et al., 2022; Aleem et al., 2024; Koleilat et al., 2024], rather than investigating how their open-vocabulary and zero-shot capacities might reshape masked image modeling itself.

In summary, although recent region-aware MIM approaches—including dynamic masking based on reconstruction loss and masking guided by external perception modules—have improved semantic focus to some extent, they remain limited by reliance on additional supervision, residual misalignment between masked regions and task semantics, and insufficient flexibility. Meanwhile, despite the strong performance of vision-language models in downstream tasks, comparatively little attention has been devoted to incorporating natural language prompts into the MIM pretraining phase as task-aware masking guidance. These observations collectively motivate the integration of vision-language semantic guidance as a means of enabling more efficient and generalizable masked image modeling.

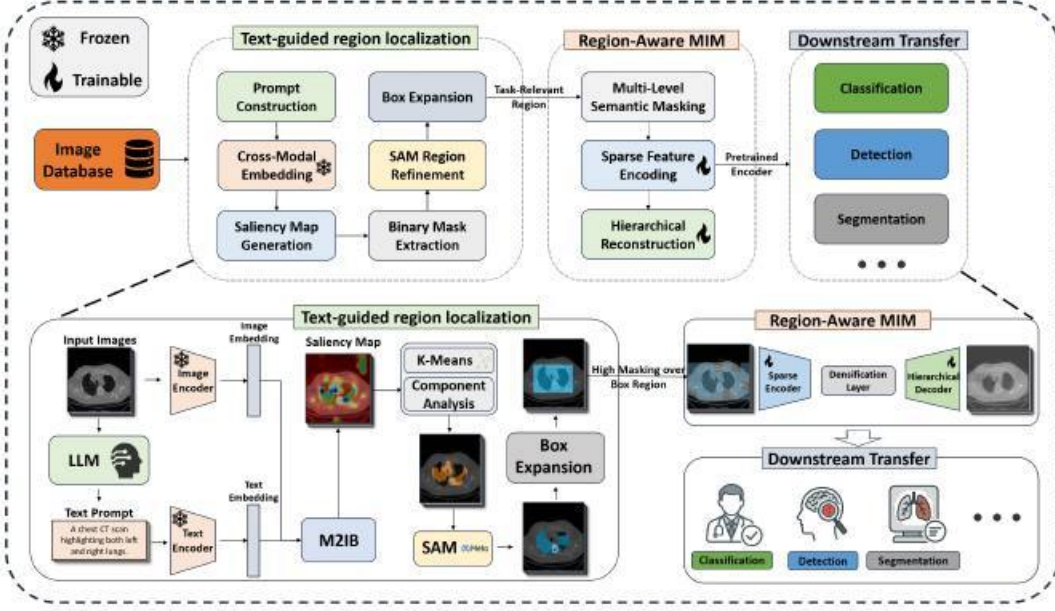


Figure 1: Overview of the MWM framework. The top panel depicts the three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel provides a magnified view of how prompts govern region localization and subsequent masking.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** is organized around two core stages: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage uses a frozen vision-language model to infer task-relevant regions from an open-vocabulary prompt, whereas the second stage assigns differentiated masking ratios across regions so that the reconstruction process is guided by semantic relevance rather than by indiscriminate spatial corruption.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a multi-stage pipeline that progressively transforms textual intent into spatial supervision. The procedure begins with LLM-based prompt generation and then uses BiomedCLIP to extract cross-modal embeddings. A saliency map is produced via the Multi-Modal Information Bottleneck (M2IB), binarized with K-Means, and refined with the Segment Anything Model (SAM). Finally, the refined mask is converted into a bounding box with a controllable expansion margin, yielding a robust spatial guide for the subsequent masked modeling stage.

Prompt Generation. In medical image analysis, textual prompt design exerts a substantial influence on the quality of region localization. To strengthen semantic guidance, large language models (LLMs) are queried with structured prompts such as “Describe the typical visual characteristics of a [category] in a [modality] image”. In addition to these full-sentence descriptions, simple category phrases are experimentally used as alternative inputs, with a detailed comparison of the resulting prompt styles provided in the experimental section.

Cross-Modal Embedding. After prompt construction, a BiomedCLIP model [Zhang et al. \(2023\)](#) encodes the medical image and its associated prompt within a shared semantic space. Given an image I and a prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These cross-modal embeddings constitute the semantic substrate on which subsequent saliency estimation is performed.

Saliency Map Generation. A cross-modal saliency estimation module based on the Multi-modal Information Bottleneck (M2IB) Wang et al. (2023b) is used to localize task-relevant visual regions by aligning image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns an importance weight to each spatial location according to its semantic relevance to the prompt. M2IB realizes this estimation by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the image and text embeddings encoded by BiomedCLIP, respectively, and γ controls the trade-off between retaining cross-modal relevance and filtering task-irrelevant visual information.

Binary Mask Extraction. After the saliency map is obtained, it is binarized with unsupervised K-Means clustering Lloyd (1982) to identify the text-guided region and to produce a preliminary foreground mask. Specifically, the map is partitioned into two pixel groups, after which the cluster with larger saliency values is designated as foreground. To further refine the mask and suppress false activations, connected component analysis is applied to the binary map, retaining only the largest foreground regions; this operation removes noisy edge responses and concentrates the guidance on core semantic structures.

Region Refinement. To improve spatial precision, SAM is incorporated as a refinement module Kirillov et al. (2023). For each selected connected region $c_i \in \mathcal{C}^*$, the minimal enclosing bounding box is computed and used as the visual prompt for SAM.

Given the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} provides a spatial structure that more faithfully conforms to the underlying anatomical or pathological region.

Box-Based Region Expansion. Because the SAM mask produced in the preceding stage may still contain boundary noise capable of perturbing the model’s focus during representation learning, the refined SAM mask is converted into a bounding box with a controllable expansion margin. This expanded box serves as the final region of interest (ROI) for downstream masked modeling and improves the robustness of the region guidance.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest produced by the localization module in Section 3.2, MWM conducts masked image modeling through a three-stage pipeline: (1) a multi-level masking strategy that assigns differentiated masking ratios according to regional importance, (2) a sparse encoder that processes only unmasked patches, and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM adopts a multi-level masking strategy that exploits the semantic regions identified by the text-driven localization process. Specifically, patches corresponding to prompt-highlighted regions, such as areas associated with tumors or organs, are masked at a high ratio so that the model is encouraged to reconstruct task-relevant semantic features. By contrast, background regions are masked lightly or left unmasked, thereby reducing redundant computation while preserving contextual support. Figure 2 presents a visual comparison of these masking patterns.

Sparse Feature Encoding. Visible patches are assembled into a sparse image and encoded according

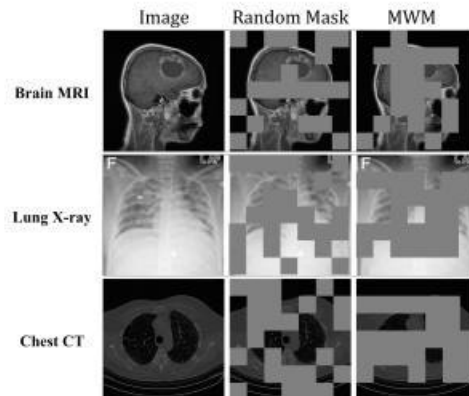


Figure 2: Comparison of masking strategies, with gray blocks denoting masked regions.

to the SparK design [Tian et al. \(2023\)](#):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ denotes the resulting multi-scale feature set. Although MWM is instantiated here with convolutional backbones, the same interface extends naturally to ViTs through token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image through a top-down procedure. At each scale l , masked locations are populated with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , after which it is upsampled and fused with features from the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is optimized with mean-squared error computed over masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ denotes the set of masked patch indices, x_i is the original patch, and $\hat{x}_i$ is its reconstruction.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Self-supervised pretraining is conducted on three datasets spanning distinct imaging modalities. (1) *Brain MRI* comprises approximately 17,000 images from Brain Tumor MRI [Cheng et al. \(2015\)](#), BRISC [Fateh et al. \(2025\)](#), and BraTS 2018 [Menze et al. \(2014\)](#), thereby covering multiple tumor categories and anatomical structures. (2) *Lung X-ray* contains roughly 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. \(2021\)](#); [Chowdhury et al. \(2020\)](#), including COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* includes 12,000 slices from a lung disease dataset [Konya \(2020\)](#), together with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation is performed on three datasets covering classification and detection-oriented tasks. (1) *Brain Tumor MRI* [Feltrin \(2023\)](#) contains 4,479 MRI images spanning 44 tumor types, including astrocytoma, glioblastoma, meningioma,

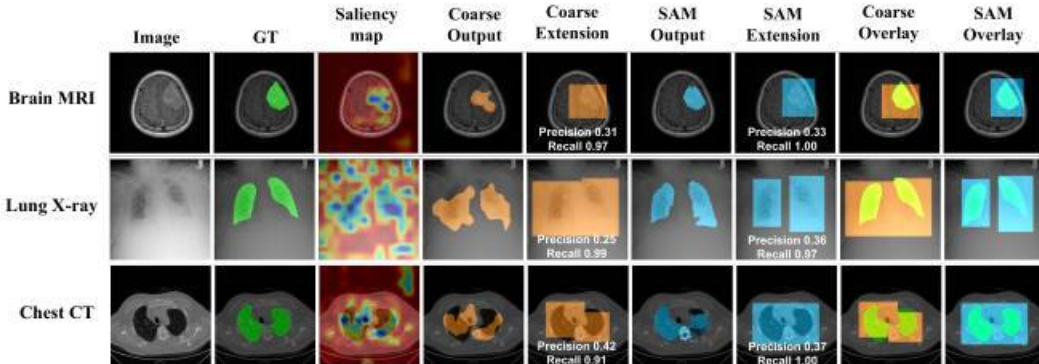


Figure 3: Representative text-guided region localization results.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance, measured by Occlusion Precision/Recall, across prompt types and refinement strategies. *Coarse Ext.* denotes bounding box extension of the coarse region; + *SAM Ext.* indicates additional refinement using SAM.

and ependymoma. (2) *Pediatric Lung X-ray* Kermay et al. (2018) consists of 5,863 pediatric chest X-rays categorized as normal or pneumonia. (3) *Chest CT Cancer* Hany (2020) comprises CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

All self-supervised methods are configured with backbones of comparable model scale: convolution-based methods use ResNet-50 He et al. (2016), whereas transformer-based methods use ViT-S Dosovitskiy et al. (2020). The input resolution is fixed at 224×224, and all models are initialized with ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} , while downstream fine-tuning preserves the same encoder and input resolution. All experiments are executed on NVIDIA Tesla A100 cards with 80 GB VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise stated, non-target variables, including prompt type and SAM usage, are held at their empirically optimal settings during evaluation to ensure controlled comparison.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first assess the effectiveness of the text-guided localization module across three imaging modalities—brain MRI, lung X-ray, and chest CT—while additionally examining how prompt type and SAM refinement influence localization quality. Specifically, predicted bounding boxes are compared with expert-annotated segmentation masks, and we report occlusion precision ($|R_p \cap R_{gt}| / |R_p|$) and occlusion recall ($|R_p \cap R_{gt}| / |R_{gt}|$), where R_p is the predicted region and R_{gt} is the ground-truth annotation.

As shown in Table 1, text-guided localization with effective prompts attains consistently high occlusion recall across all datasets (around 0.82) while maintaining reasonable precision. This result confirms that the framework can identify task-relevant regions using textual descriptions alone, even in the presence of modality-specific anatomical variation (see Figure 3 for visualized examples).

Effect of Prompt Type on Region Localization. To further quantify the role of language design, we evaluate two prompt types with respect to localization performance. As reported in Table 1, descriptive sentence prompts yield stronger results for Brain MRI and Chest CT, suggesting that richer semantic specification improves localization. In contrast, concise phrases obtain higher recall for lung radiographs (0.82 vs. 0.66), plausibly because the broad anatomical extent of the lungs makes general lexical cues sufficiently informative.

Effect of SAM Refinement on Region Localization. We also examine the contribution of spatial refinement by applying SAM to the coarse masks. As shown in Table 1, SAM refinement improves precision while preserving high recall; for example, in Brain MRI, precision increases from 0.22 to 0.30 while recall remains above 0.94. This pattern demonstrates the utility of SAM for converting coarse localization outputs into more spatially precise guidance.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next evaluate the generalization and representational capacity of MWM by comparing it with representative self-supervised learning (SSL) methods from both reconstruction-based and contrastive learning paradigms. To support fair comparison, all methods are pretrained and evaluated under consistent datasets and training settings.

Image Classification. We examine classification performance on three downstream medical imaging datasets. Following standard practice, results are reported under two evaluation settings: (1) *full fine-tuning*, in which the entire model is updated, and (2) *linear probing*, in which the encoder is frozen and only a linear classifier is trained on top.

- **Full fine-tuning:** Table 2 summarizes classification accuracy for MWM and several representative self-supervised methods, including MAE He et al. (2022), SparK Tian et al. (2023), AnatoMask Li et al. (2024), SimCLR Chen et al. (2020a), MoCoV2 Chen et al. (2020b), and BYOL Grill et al. (2020). MWM consistently achieves the highest performance, thereby demonstrating strong generalization across heterogeneous imaging modalities. The largest improvement appears on the Chest CT dataset, where MWM reaches 97.5% accuracy, exceeding SparK by +3.1 and AnatoMask by +1.7. This improvement underscores the advantage of text-guided masking, which emphasizes semantically critical yet spatially sparse regions, such as lung lobes in CT scans, while avoiding redundant masking of less informative areas.
- **Linear probing:** To further assess the quality of the learned representations, we conduct linear probing experiments. For simplicity, MWM is compared with two recent state-of-the-art masked image modeling methods, SparK and AnatoMask. As shown in Table 3 MWM substantially outperforms both methods across all datasets, indicating that text-driven pretraining yields features with stronger linear separability.

Object Detection and Instance Segmentation. To assess whether the learned representations generalize beyond classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset Hamada (2025). As shown in Table 4 MWM obtains the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), exceeding SparK by +1.3 and +1.1 points, respectively. These results indicate that MWM transfers effectively to both object-level localization and pixel-wise segmentation.

In summary, incorporating text-driven masking during pretraining enables the model to acquire more informative and transferable representations. The resulting semantic guidance consistently improves classification, detection, and segmentation performance, underscoring its effectiveness for strengthening visual representation learning.

4.5 ABLATION STUDY

We further conduct ablation experiments to isolate the individual contributions of prompt design, SAM-based region refinement, and masking ratio to downstream classification performance in MWM.

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Fine-tuning classification accuracy (%) on downstream datasets, where MIM denotes Masked Image Modeling and CL denotes Contrastive Learning.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on downstream datasets with the encoder kept frozen.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) performance (%) on the BR35H dataset.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As shown in Table 5, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, whereas the opposite pattern is observed on lung X-ray. This finding is consistent with the localization results in Section 4.3, where prompts that produced more accurate region identification also yielded stronger downstream performance. Importantly, both prompt types, irrespective of granularity, consistently outperform random masking across all datasets. This result confirms that text-guided masking facilitates semantic alignment between pretraining and downstream tasks, a property that is crucial for improving self-supervised learning performance. In addition, the SAM ablation shows that incorporating this module during pretraining consistently improves downstream classification, thereby confirming its effectiveness in strengthening text-guided masking.

4.5.2 EFFECT OF MASKING RATIO

As shown in Figure 4, MWM achieves 96.8% classification accuracy at a masking ratio of only 40%, a setting substantially lower than the approximately 70% ratio commonly used in masked image modeling. Despite this reduced corruption level, it exceeds SparK’s best performance at 70% masking by +2.4% in absolute accuracy. To exclude the possibility that SparK’s weaker performance results from a suboptimal masking configuration, SparK is additionally evaluated under 40%, 50%, 60%, and 70% masking ratios. The results confirm that 70% masking produces SparK’s highest accuracy, thereby supporting the robustness of the comparison. These findings challenge the prevailing assumption that high masking ratios are intrinsically optimal for masked image modeling. Instead, they emphasize the advantage of semantically guided, task-aware masking strategies such as MWM, which achieve stronger downstream performance at lower masking ratios by preserving task-relevant content while reducing redundancy.

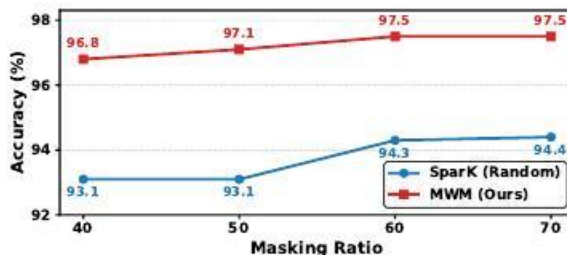


Figure 4: Classification performance on the Chest CT dataset under alternative masking strategies and masking ratios.

5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that integrates text-guided semantic localization with region-aware masking. By leveraging natural language prompts to identify task-relevant areas and by applying differentiated masking to those re-

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under different prompt types and region refinement strategies across three imaging modalities. *No Prompt* refers to random masking without text-guided localization. *w/* and *w/o* denote configurations with and without SAM, respectively.

regions, MWM enables semantically aligned representation learning without requiring per-image annotations.

Comprehensive experiments substantiate both the effectiveness and the generalizability of MWM: (1) natural language prompts reliably guide semantic region localization, thereby validating the feasibility of text-driven masking; (2) MWM consistently outperforms existing methods (e.g., SparK, AnatoMask) in classification across three imaging modalities and also yields improvements in detection and instance segmentation; (3) MWM preserves strong performance even at lower masking ratios (e.g., 40%), underscoring the value of semantic guidance in self-supervised pretraining.

Limitations. Despite these promising results, the framework has several limitations. First, although text prompts provide flexible guidance, the robustness of MWM to variations in prompt style, vocabulary, or noise has not yet been systematically examined. Second, reliance on external textual descriptions introduces an additional dependency that may weaken generalization in scenarios where reliable prompts are unavailable, ambiguous, or inconsistent.

Looking forward, text-driven, region-aware masking offers a principled path toward more semantically grounded self-supervised learning, and extending MWM beyond medical imaging may create new opportunities for broader vision–language pretraining.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

LLMs were used *only* for language editing and presentation polishing, including grammar, phrasing, and minor stylistic clarification of text written by the authors. They were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.